Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/09/2023
PI Name:	Cornforth, Michael Ph.D.		
Project Title:	Molecular Characterization of Transmissible Chromosome Aberrations Produced by Ions of Intermediate and High Atomic Number: grant 80NSSC21K0679		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Update per the Principal Investigator (PI): Yongjia Yu, Ph.D. is no longer with the project as CoInvestigator (Ed, 3/22/23).		
COI Name (Institution):			
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	NOTE: Continuation as directed research of "Molecular Characterization of Transmissible Chromosome Aberrations Produced by Ions of Intermediate and High Atomic Number," grant NNX15AG74G; Principal Investigator Dr. Michael Cornforth continues as PI. Risks associated with crewed space flight involve extremely low fluences of mixed-field galactic cosmic radiation (GCR), and secondary neutrons formed through the interaction of lighter ions with the structure of spacecraft. It is simply not possible to duplicate this environment in any biologically meaningful experimental setting on Earth. The best solution to this problem is to simulate key aspects of the deep space environment by focusing on “representative” aspects of GCR. For obvious reasons, it is not feasible to measure directly biological risk to astronauts. Thus, we are forced to employ experimental model systems that mimic central aspects of the deep space radiation environment, and		

Task Description:	which employ surrogate biological endpoints of principle relevancy to biological endpoints of concern, like cancer. Most all ionizing radiation (IR)-induced chromosomal aberrations are exchanges that occur between different chromosome (e.g., translocations), or between regions within the same chromosome (e.g., inversions). Beyond the basic investigative platform that it provides to the science of radiation biology, knowledge concerning the genesis of IR-induced breakpoint junctions carries specific potential benefits. As for NASA and its priorities are concerned, a more detailed view of the enzymology of the rejoining reactions themselves would be of immense value to modelers, particularly as it relates to ultra-low dose risk assessment. Another benefit lies in the prospect that changes in radiation quality will influence the molecular structure of IR-induced breakpoint junctions. Colonies derived from the single-cell expansion of irradiated human telomerase reverse transcriptase (hTERT)-immortalized BJ-1 fibroblast cultures form the basis of this system. These cytogenetically characterized clones form the starting material for subsequent sequencing analysis. Specific Aims Aim 1: Determine if analysis of IR-induced chromosome aberration (CA) breakpoint junctions suggest different molecular pathways are involved, with respect to high- versus low Linear Energy Transfer (LET) exposure. Aim 2: Determine if the sizes of co-deleted sequences at breakpoint junctions differ for those produced by high versus low LET IR; these may form the basis of an LET-dependent biomarker. Aim 3: Determine if the breakpoints associated with complex chromosomal exchanges are different than those associated with simple reciprocal exchanges that involve only two chromosomes. Aim 4: Examine the breakpoint junctions of exchange aberrations induced by medium-to-high energy neutrons. Aim 5: Characterize molecular differences between the breakpoints produced by different ultra-low dose ions.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This study represents a continuation of Chromosome Aberrations I (NNX15AG74G) project; the objectives and research goals remain largely unchanged. We seek fundamental information at the molecular level about the nature of chromosome aberration breakpoints induced by ionizing radiations (IR) that differ vastly in ionization density, comparing terrestrial gamma rays to those of much higher linear energy transfer rates. To accomplish this will require further development and refinement of an experimental model system for which we have previously established proof-of-principle efficacy. This includes implementation of advanced methods of massively parallel sequencing such as single molecule/real-time sequencing.
Research Impact/Earth Benefits:	Radiation-induced reciprocal chromosome translocations and inversions are particularly important in that regard as they relate to crewed space activities. In addition to causing cancer, their appearance also accompanies ongoing genome instability processes associated with their progression. The fact that these particular chromosome aberrations are transmissible (non-lethal) also makes them ideal candidate biomarkers of accumulated radiation exposure. We argue that molecular analysis of breakpoint junctions formed as the result of translocations and inversions is vital to understanding the process of exchange aberration formation, since it is here where underlying repair/misrepair pathways leave their “molecular fingerprints.” Regarding relevance to NASA’s concerns, the study of chromosome aberrations stands to tell us much about mechanisms underlying the cancer process itself. The relationship between particle energy/track structure and radiogenic changes to the genome represents an important first step in understanding 1) basic dose-response relationships at low fluences and 2) fundamental carcinogenic processes that may ultimately form the basis for subsequent mitigation strategies.
Task Progress:	The overall structure of this proposal may be described as follows. For Objective 1, single human cells, initially exposed to various radiations, were grown into colonies that were later randomly isolated and cryopreserved for subsequent analyses. The first stage of the analyses involved 24-color chromosome painting (mFISH) in identifying those clones whose cells harbored microscopically visible chromosome rearrangements such as translocations. These clones constituted the starting material for use in Objective 2, which involved the DNA sequencing of breakpoint junctions associated with radiation-induced translocations and inversions. We proposed that DNA sequencing of these breakpoints would likely uncover fundamental differences in the types of damage produced by common gamma rays (and their underlying repair processes) compared to that produced by radiations encountered in deep space environments (HZE particles), specifically high energy 56Fe accelerated ions produced at NASA’s NASA Space Radiation Laboratory (NSRL) Brookhaven facility. Updated progress includes a sort of meta-analysis of data related to Objective 1, wherein we examined individual cells at their first metaphase following exposure to determine the likelihood of each being either lethal or nonlethal (transmissible). On the basis of such cell-by-cell analysis, we were able to reconstruct a dose-response for the frequency of only viable/transmissible cells that also contained aberrations visible by mFISH. This was done for cells exposed to gamma rays and for cells exposed to 56Fe heavy ions. In both cases, the dose responses exhibited concave-downward curvature that produced a crescent-shaped dose response. When the two curves were compared against one another, we found that the relative potency of heavy ions was scarcely higher than that of gamma rays. In fact, after modeling the two-dose responses, we found that the effectiveness of heavy ions was expected to be actually less effective per unit dose than gamma rays at higher doses. Following the publication of these results (Radiat Res. in press), we will consider Objective 1 to have been completed. The majority of progress of this report centers on Objective 2 and the sequencing of chromosome aberration exchange breakpoints in cells clonally expanded from a single progenitor cell exposed to either gamma rays or 1.0 GeV accelerated 56Fe ions. We had earlier determined that various short-read sequencing approaches were insufficient for our purposes due to the ubiquitous presence of repetitive DNA sequences in the human genome. As such, we proposed the use of various newer sequencing strategies designed to circumvent this problem, among them mate-pair sequencing and single-molecule/real-time sequencing (SMRT). The latter proved especially useful for our purposes and has allowed for the breakpoint sequencing of several additional clones. Our data to date has not revealed obvious systemic differences in the nucleotide structure surrounding the breakpoint junctions produced in clones exposed to the two radiation types. All appear to be compatible with a nonhomologous

	end-joining enzymatic pathway that involves relatively short stretches of microhomology. However, we have not fully sequenced enough clones to conclude this point. We have several additional clones in the pipeline which are undergoing various stages in the sequencing process using SMRT. Analysis of additional clones will constitute our major effort going forward, with the first objective to publish our sequencing findings for a representative sample of twelve clones, six each from the 137Cs-gamma ray and 56Fe-exposed cohorts. While adequate progress is being made using SMRT, we are optimistic that a newer ion-pore sequencing approach developed by Oxford Nanopore will eventually be the best path forward for these studies, particularly so in combination with an integrated real-time algorithmic platform (Readfish) that allows for so-called selective sequencing. This is part of an augmentation grant to our main proposal through my graduate Student, Stephen Kunkel [Adaptive nanopore sequencing to increase the sensitivity of detecting transmissible chromosomal aberrations induced by ionizing radiation]. The selective sequencing approach offers distinct advantages over SMRT, such as lower cost and allowing for even longer read lengths than SMRT. To our knowledge, selective sequencing has never been applied to the analysis of radiation-induced exchange breakpoint analysis. Consequently, we have encountered specific challenges in its implementation that we are currently working to overcome.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Cornforth M, Loucas B, Shuryak I. "Dose-dependent transmissibility of chromosome aberrations in human lymphocytes at first mitosis. II. Biological effectiveness of heavy charged particles versus gamma rays." Radiat Res 2023 Jan 17;199(3):283-89. https://doi.org/10.1667/RADE-22-00141.1 ; PMID: 36648766 , Jan-2023
Articles in Peer-reviewed Journals	Loucas BD, Shuryak I, Kunkel SR, Cornforth MN. "Dose-dependent transmissibility of chromosome aberrations at first mitosis after exposure to gamma rays. I. Modeling and implications related to risk assessment." Radiat Res 2022 Jan 17; 197(3): 376-83. https://doi.org/10.1667/RADE-21-00180.1 ; PubMed PMID: 35030259; PubMed Central PMCID: PMC9109216 , Jan-2022
Articles in Peer-reviewed Journals	Cornforth M. "Commemoration of JRK Savage, scientist extraordinaire." Radiat Res 2022 Sep 26; 198(4):444-45. https://doi.org/10.1667/RADE-22-0JRKS.1 ; PubMed PMID: 35943869 , Sep-2022